

Yingrui Hou

Scientific Data Analyst | Python, Statistical Modelling & Reproducible Workflows

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SUMMARY

Scientific data analyst with a PhD in experimental physics and several years of experience developing Python/C++ workflows for statistical modelling, data manipulation, calibration, validation, inference, and monitoring on large, noisy, real-world measurement datasets.

Experienced in building reusable and reviewable analytical pipelines, producing documented figures and statistical summaries, validating ML and inference workflows, and collaborating with scientists in international teams.

Motivated to transfer this rigorous open-source programming practice to clinical-trial analytics while developing expertise in CDISC standards and clinical development.

CORE SKILLS

- Data analysis: Python, NumPy, pandas, data manipulation, exploratory analysis, statistical summaries, visualisation
- Statistical modelling: maximum-likelihood estimation, probabilistic and mixture models, joint fitting, latent-parameter inference
- AI and ML: scikit-learn, statsmodels, CatBoost / Gradient Boosting, feature studies, train/test diagnostics
- Validation and QC: model validation, calibration, bias correction, uncertainty checks, pseudoexperiments, anomaly diagnosis
- Reproducible programming: Git, code review, modular workflows, CI/CD practices, YAML / Shell / CMake automation
- Engineering and delivery: C/C++, Linux, shared codebases, large-scale data processing, technical documentation

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Developed reusable Python/C++/ROOT pipelines for data preparation, exploratory analysis, statistical inference, validation, and production of documented analytical outputs on large datasets.
- Implemented probabilistic and maximum-likelihood models, including joint fits, under background contamination, finite resolution, incomplete event-level ground truth, and dataset mismatch.
- Built CatBoost-based workflows with feature studies, train/test diagnostics, threshold scans, reproducible model application, and gradient-boosting reweighting.
- Developed automated monitoring and QC workflows that compared expected and observed behaviour, detected anomalies, and traced discrepancies to calibration, detector, or software causes.
- Contributed covariance-aware checks, pseudoexperiments, uncertainty-stress studies, and control-sample validation for advanced inference workflows.
- Produced structured statistical summaries, diagnostic figures, documented methods, and interpretable results for review by multidisciplinary collaborators.
- Worked in Linux-based shared codebases using Git, modular programming, collaborative review, reproducible configurations, and automated workflows across international teams.

Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed Geant4/C++ simulation and statistical-analysis workflows combining response modelling, calibration, signal reconstruction, bias-aware interpretation, and validation against experimental data.

SELECTED OUTCOMES

- Improved reconstructed-energy resolution by about 2x through time-resolved component modelling, statistical fitting, and calibration on realistic measurement data.
- Identified a reconstruction-software issue through structured validation and monitoring; the issue was subsequently corrected downstream.

SELECTED PROJECTS

Advanced Inference Workflow Validation

Contributed the proof-of-principle validation needed to move a new basis-expansion inference idea from theory toward a usable workflow under realistic measurement conditions.

Methods: basis expansion, covariance-aware joint fitting, pseudoexperiment validation, statistical background subtraction

Weak-Signal Inference

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

Methods: likelihood fitting, control-sample validation, production yield workflows

Domain Adaptation for ML

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting

Time-Series Inference

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting

Test-Data Validation & Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

EDUCATION

- Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021
- BSc in Applied Physics - China University of Mining and Technology, 2010-2014
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2